

Experimental and numerical study on the thermal environment of a data center

Yu Tang^{a,b}, Xuelian Bai^{a,b,*}, Chaoqiang Jin^{a,b,c}, Yusen Fu^{a,b}, Wangxin Mao^{a,b}, Xin Xu^{a,b}

^a School of Civil Engineering, Chongqing University, Chongqing, 400045, China

^b National Centre for International Research of Low-carbon and Green Buildings, Chongqing University, Chongqing, 400045, China

^c GLP Purui Data Technology (Shanghai) Co., Ltd, Shanghai, 200135, China

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ABSTRACT

Data centers are energy-intensive facilities, and a suitable thermal environment is vital for the safe and reliable operation of information technology equipment and energy saving. In this case study, field experiments and thermal environment evaluation were conducted in an existing data center. However, some problems were reported, including the relative humidity of the rack inlet air exceeding 60%, hot-cold air mixing, non-uniform temperature distribution, local overheating, and a low average temperature difference between the return and supply air of computing room air conditioning units (CRACs). To overcome these dilemmas, three cost-efficient and convenient methods for improving the thermal environment were proposed, including installing blanking panels, closing unnecessary perforated tiles, and increasing the set return air temperature of CRACs. The temperature field and flow field were simulated using the software 6SigmaRoom to predict the effects of the optimization methods. Finally, after the optimization scheme was implemented, field experiments were performed again to examine the application effects of optimization measures. The results demonstrate that the temperature distribution was more uniform, and the thermal environment was considerably improved. Meanwhile, the average temperature difference between the return and supply air of CRACs increased from 6.7 °C to 8.8 °C, resulting in a decrease in the supply air volume of CRACs of about 10903.37 m³/h, and saving the energy consumption of the fans around 96 kWh/day.

1. Introduction

Data centers (DCs) refer to the place housing information technology equipment (ITE) and telecommunication equipment in standard-sized racks [1]. With the increasing requirements for high-speed data processing, the energy consumption issue of DCs has become increasingly prominent. The energy consumption of DCs accounted for 1.3% of the energy use worldwide in 2012. In addition, the ratio reached up to 3% in 2017 and has been predicted to increase to 4.5% by 2025 [2,3]. The cooling system accounts for approximately 30%–50% of the total energy consumption of DCs [4]. Therefore, there is a significant need to reduce the energy consumption of DCs by adopting efficient cooling technologies. In addition, the thermal environment is extremely important for the safe and reliable operation of the ITE and energy saving in DCs. When operating at a high temperature for a long period of time, the failure rate of the ITE increases, the lifetime decreases [5], and crashes

may occur [6]. The high relative humidity (RH) may also damage the ITE, such as through a conductive anode failure, hygroscopic dust failure, tape media failure, or corrosion [7]. When the ITE is exposed to a low RH (<30%) for a long period of time, various problems may occur, such as friction electrostatic discharge and the breakdown and short circuits of the components [8]. The fourth edition of *Thermal Guideline for Data Processing Environments* published by ASHRAE in 2015 reflects the general requirements of ITE for indoor thermal parameters, emphasizing the guarantee of the equipment reliability and performance [9]. The recommended range of the dry-bulb (DB) temperature is 18 °C–27 °C, and the humidity range is –9 °C–15 °C dew-point (DP) and 60% RH. In the currently available Chinese *Code for the Design of Data Centers*, the recommended range of DB temperature is consistent with ASHRAE, and the humidity range is 5.5 °C–15 °C DP and 60% RH. Therefore, it is essential to save energy while improving the thermal environment in DCs by applying efficient cooling strategies.

Numerous researchers have explored various cooling technologies to

* Corresponding author. National Centre for International Research of Low-carbon and Green Buildings, School of Civil Engineering, Chongqing University, Chongqing, 400045, China.

E-mail address: xuelianbai@cqu.edu.cn (X. Bai).

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Nomenclature			
T_{in}^r	IAT of the rack, °C	DCs	Data centers
T_{out}^r	OAT of the rack, °C	ITE	Information technology equipment
T_{sup}^C	SAT of the CRAC, °C	RH	Relative humidity
T_{ret}^C	RAT of the CRAC, °C	DB	Dry-bulb
L_{CRAC}	Air volume of the CRAC, m ³ /h	DP	Dew-point
P_{CRAC}	Power consumption of the CRAC, kW	SAT	Supply air temperature
R^2	Coefficient of determination quantifying goodness of fit, –	RAT	Return air temperature
<i>Greek symbols</i>		CAC	Cold aisle containment
θ	Fan speed of the CRAC, %	HAC	Hot aisle containment
k	Turbulent kinetic energy, m ² /s ²	IAT	Inlet air temperature
ϵ	Turbulent kinetic energy dissipation, m ² /s ²	OAT	Outlet air temperature
<i>Subscripts</i>		SHI	Supply heat index
i	The i th rack in a row	RTI	Return temperature index
j	The j th row in the DC	FBs	Flexible baffles
<i>Abbreviations</i>		IBFs	Induction bypass fans
CRACs	Computing room air conditioning units	IRCA	In-rack cold aisle
		CFD	Computational Fluid Dynamics
		DDM	Data-driven model
		PDU	Power distribution units

improve the thermal environment and alleviate energy consumption in DCs, which can be divided into liquid cooling, free cooling, airflow management, and adjusting the CARC parameter settings. Liquid cooling has been applied to large-scale DCs with high heat dissipation density [10–12]. However, the traditional air-cooled method remains the mainstream which has been well developed and widely used. Free cooling, which uses natural cold resources to cool the DCs, is one of the best choices for reducing cooling energy consumption [4,13–15]. However, the application of free cooling has certain limitations because it needs to meet the corresponding climate and resource conditions, such as sufficiently dry and cold outdoor air and adjacent lakes or oceans with high-quality water. Compared with free cooling, adjustments of the CARC parameters and airflow management are believed to be effective and convenient strategies. For CRACs, the optimum range of operating parameters for energy saving should guarantee the safe and reliable operation of ITE. Generally speaking, a lower supply air temperature (SAT) and a higher supply air volume can improve the thermal uniformity and relieve local hotspots. However, it was reported that the operating strategy of a high SAT and high air volume performed better under the same cooling capacity [16,17]. It was estimated that the cooling energy conservation ratio was 4.3%–23.9% for every 1 °C increase in the SAT. The higher the ambient temperature is, the more energy can be saved [18]. According to Oró et al. [19], combining the air distribution optimization with an increase in the SAT and a decrease in the air volume can effectively alleviate the cooling energy consumption in a DC. These studies mostly focused on the impact of adjusting the SAT on the thermal performance and energy consumption and have paid little attention to the effects of changing the RAT set points.

Airflow management refers to providing a uniform airflow, eliminating hotspots, and improving the thermal performance and cooling efficiency with minimum energy consumption. It includes preventing bypass and recirculation [20–26] and improving the airflow uniformity [16,17,27–45]. The strategies for preventing air mixing include cold aisle containment (CAC), hot aisle containment (HAC), and installation of blanking panels. Compared with the semi-CAC (with sides), the full CAC (with roofs and sides) can significantly improve the thermal environment, especially in DCs with high power density [20]. Moreover, applying a full CAC has shown a significant enhancement in the temperature distribution uniformity at the rack inlets whenever cold air is under- or over-provided [21]. Through an experimental analysis [22], the maximum rack inlet air temperature (IAT) with CAC was 3 °C lower

than that with HAC under the same heat load and air-conditioning environment. At the same time, the fan power with CAC was half that of HAC under the same IAT of the ITE. According to Refs. [23,24], HAC was more efficient than CAC because HAC allowed hotter air to directly return to the CRACs and the setpoints of the supply air could be higher. HAC is a better choice for a new DC, and CAC is easier and less costly to implement for existing room-based cooling. In practice, containment systems cannot guarantee an ideal airtightness. With an operating condition of high air leakage rate, HAC exhibited a better thermal performance than CAC [25]. In addition, applying blanking panels is important for preventing the local bypass and recirculation, particularly for DCs employing aisle containment [26].

For the raised-floor DCs, cold air supplied from the CRACs passes through the raised-floor plenum into the cold aisles and is then drawn by fans to cool the servers in racks. It is effective to improve the thermal environment and cooling efficiency by optimizing the arrangement of equipment (CRACs, racks, and servers) [28–31] and improving the uniformity of static pressure and airflow distribution in the plenum [17, 29,32–41]. The CRACs can be parallel or perpendicular to the cold aisle. Compared with the former, the latter can provide more even airflow along the cold aisle and reduce the cold-hot air mixing [27]. Placing CRACs at the ends of hot aisles rather than cold aisles resulted in a lower IAT of racks and a better thermal environment [28], particularly for high-density rack rows [29]. This resulted from the fact that more cooling air entered into the cold aisle and the hot exhausted air returned to the CRACs with a smaller distance. Compared with the other server arrangements (discrete, clustered, and segmented), the best thermal performance and cooling effects were obtained when racks were uniformly power-loaded [29,30]. The best supply heat index (SHI) and rack temperature distribution can be achieved when applying double servers with double voids [31].

Numerous researchers have reported that the plenum height [17, 32–34], perforated tile open area [29,33,35], and obstructions [35–37] have a significant impact on the flow field in the raised-floor plenum. Although raising the plenum height can improve the airflow distribution and cooling efficiency, there was no noticeable benefit when it exceeded the threshold value of approximately 0.61 m (24 in.) [17,32,33]. In addition, the recommended range was 1.0–1.2 m for open aisle, 0.6–0.8 m for CAC, and 0.4–0.6 m for HAC [34]. Similarly, a threshold value exists for the open area of perforated tiles for the balance between the flow resistance and airflow uniformity. A 25% opening was regarded as

the best option for presenting a uniform airflow rate along the rack row and minimal heat accumulation at the racks [29,33,35]. Fulpagare et al. [37] pointed out that obstructions (e.g., water pipes and electrical circuits) reduced the airflow rate by up to 80% and increased the tile outlet air temperature by 2.5 °C, increasing the local overheating phenomenon. Bhopate et al. [36] proposed optimized obstruction arrangements to minimize the detrimental effect on the thermal performance and presented guidelines for redisposing existing obstructions and installing new obstructions. Fakhim et al. [35] found that locating obstructions under hot aisles instead of cold aisles resulted in a more even airflow rate. In addition, in recent years, some optimization methods addressing the non-uniform pressure field and flow field in the plenum have been proposed, which include adding beneficial obstructions [17,38,39] and applying enhanced facilities [40,41]. Applying inclined partitions in the plenum can enhance the airflow uniformity along the rack row and decrease the intake and exhaust air temperature by as much as 1.7 °C–2.9 °C [17]. Yuan et al. [38] numerically investigated the improvement in the rack outlet air temperature (OAT) distribution by installing flexible baffles (FBs) underneath the raised floor in a DC with CAC. The results demonstrated that adopting 45° FBs can provide the best thermal performance. According to Tradat et al. [39], applying porous obstructions in the plenum can eliminate vortices under the raised floor and thus provide a more even airflow, improving the rack IAT distribution. By using partitions with the appropriate geometry, the leakage was reduced by half, the SHI was reduced by 10.4%, and the percentage of ITE that exceeded the ASHRAE recommended SAT was decreased by 40%. Erden et al. [40] proposed the installation of induction bypass fans (IBFs) beneath the cold aisles in a DC with CAC to improve the cooling efficiency. The results showed that this method can reduce the fan power of CRACs by as much as 52% while improving the airflow uniformity. Similarly, Song et al. [41] presented a numerical investigation of the airflow and thermal characteristics within a contained DC with fan-assisted perforations. The results indicated that using the straightened tile flow significantly reduced the hot air recirculation at the bottom region of the racks, thus improving the cooling performance.

Moreover, innovative airflow optimization methods have recently been proposed. A curved air guide sleeve at a depth of 0.5 m was installed at the CRAC outlet with the opening oriented toward the rack, which made the supply airflow to the cold aisle more orderly and fully utilizes the cooling capacity [42]. Yuan et al. [43,44] conducted experiments and simulations to analyze the effect of employing FBs with different sizes and angles to the rack outlet and inlet on the thermal environment in a high-power DC, respectively. They concluded that applying appropriate FBs could significantly reduce the local overheating and improve the air distribution and cooling efficiency. In 2020, Yuan et al. [45] explored the air optimization effects of adopting an in-rack cold aisle (IRCA) to represent the traditional CAC in a high-density raised-floor DC through Computational Fluid Dynamics (CFD) models verified by experiment results. It was concluded that the optimal thermal environment appeared when a 0.6 m × 0.6 m IRCA and partition were used. Song et al. [16] numerically and experimentally analyzed the effects of employing a deflector in the cold aisle to improve the air distribution. The results showed that this method can reduce the heat accumulation and improve the cooling efficiency and that applying a convex deflector with a 1.7-m height performed the optimum thermal environment. In recent years, employing statistical or machine-learning-based tools for efficient airflow management and thermal management has increasingly become a focus of managers and researchers. Athavale et al. [46] compared the performance of three different data-driven model (DDM) methods in predicting both the steady-state and transient rack IAT distributions in DCs. An increasing number of researchers have been devoted to developing fast, adaptive, and accurate hybrid models for predicting spatiotemporal temperature changes inside a DC and hence achieving an optimized airflow distribution [47,48].

Up till now, numerous studies have been performed on the

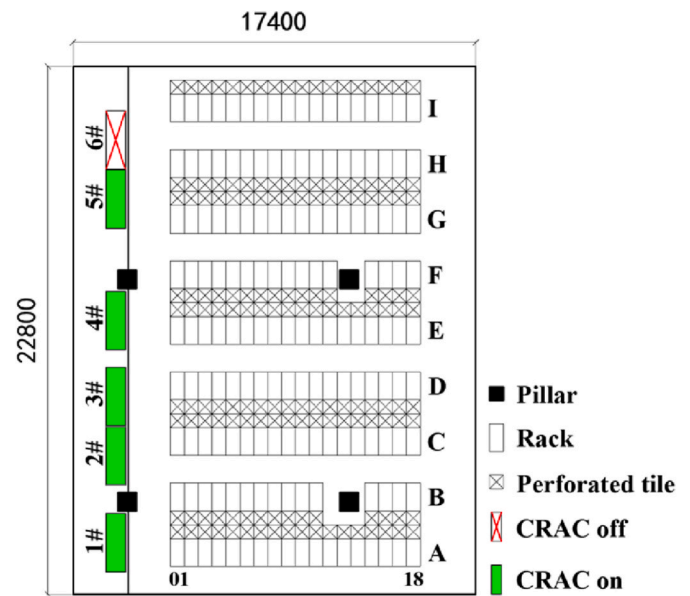


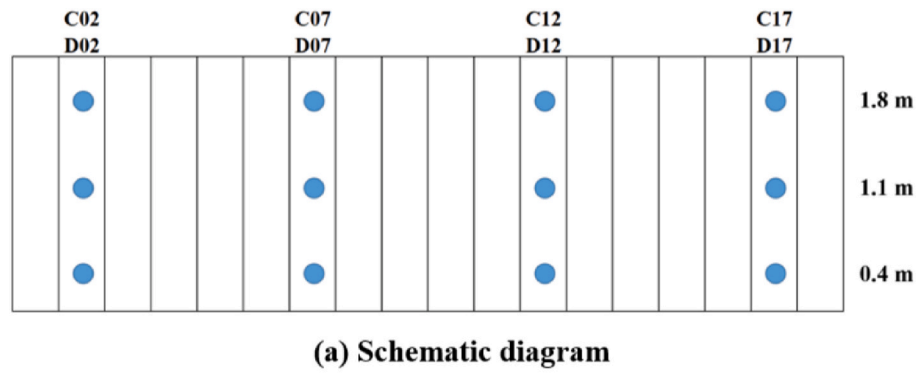
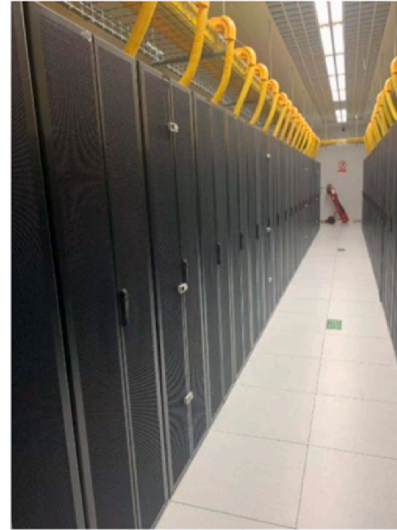
Fig. 1. Layout of the studied DC.

influencing factors and optimization of the thermal environment. However, portions of improvement methodologies work effectively during the design stage but are inapplicable to the existing DCs when considering the cost and maneuverability, such as the arrangement of CRACs, plenum configuration, IBFs, and HAC. The owners and managers of existing DCs prefer the measures which are cost-effective and easily implemented. Moreover, improvement technologies usually present various effectiveness relying on the operating conditions. For this purpose, several technical measures are simultaneously implemented, and their mutually influential effects should be paid attention to carefully. Furthermore, many optimization research works have only presented simulation results, and the practical effects of optimization measures have not been verified or checked by the field experiment data of the optimized or retrofitted DC. In addition, due to the complex conditions of the site, most measures are partially implemented, and thus the application effectiveness of these measures cannot achieve the expected targets. Thereby, how to improve the thermal environment of existing DCs is worthy of being examined constantly. This case study focuses on the effects of the simultaneous application of multiple measures in a real engineering project. The remainder of this paper proceeds is structured as follows. Firstly, field experiments were conducted to investigate the thermal environment in an existing DC. Then, the CFD models, which were in good agreement with experiment results, were applied to conduct optimization research works. Finally, the application effects of the optimization scheme were analyzed through on-site measurements.

2. Field experiment and evaluation on thermal environment

2.1. Data center description

Fig. 1 shows the layout of the studied DC. The dimensions of the DC were 22.8 m (L) × 17.4 m (W) × 6.9 m (H), and the CRAC and IT regions were separated by an inner wall. A raised-floor air supply and ceiling plenum air return system was applied in the DC, and the height of the raised floor plenum and that between the ceiling and raised floor were 1.1 m and 3.2 m, respectively. The dimensions of the perforated tiles were 0.6 m × 0.6 m. In addition, the cold aisles were fully contained. There were 157 server racks arranged in nine rows, the design power of which was 4.5 kW per rack, and the rows were marked A through I. The actual total IT load was 291.46 kW, and the servers were mainly clustered in rows A through D. Six CRACs were used in the DC to provide

**(b) Front door****(c) Rear door****Fig. 2.** Measurement points of racks.

cooling air, and the rated cooling capacity and airflow rate were 150 kW and 9000 l/s, respectively. The six CRACs were marked 1# to 6#, and all CRACs operated throughout the entire day except for a standby one. Additionally, the CRAC independent return air temperature control strategy was applied to the DC, and the set return air temperature (RAT) and RH were 25 °C and 50%, respectively.

2.2. Field experiment description

The field experiment was carried out from August 13 to 15, 2018. During the measurement, the output server power and working conditions generally remained stable. The population, arrangement, and rated power of the servers were counted, and the real-time power of the racks was collected from power distribution units (PDUs) every 15 min. Considering that the total server power in rows C and D was the highest and that conducting field measurements on a large scale was restricted by the owner, racks in rows C and D were selected as the test objects. The inlet and outlet air temperature and RH of the racks were acquired with a 30-s interval using the temperature-humidity recorders, and the measurement points were 0.4, 1.1, and 1.8 m high from the raised floor, respectively, as shown in Fig. 2. A temperature-humidity recorder was mounted at the center of every CRAC outlet to record the air temperature with a 30-s interval, as presented in Fig. 3. The RAT and fan speed of each CRAC were obtained four times per hour from the monitoring system. In addition, field experiments about the air volume and power consumption of the CRAC are conducted to obtain its actual operating condition. Changing the fan speed from 60% to 100% with a 1% interval, the real-time power consumption was acquired from PDUs.

Changing the fan speed from 60% to 100% with a 5% interval, the real-time air volume can be obtained by using hot-wire anemometer to acquire the airflow velocity of 75 measurement points uniformly distributed on the CRAC return air area. A flow hood was used to collect the outlet air temperature and volume across the perforated tiles, as shown in Fig. 4. Besides, an infrared thermal imager was employed to show the air temperature distribution at the rack outlet. The specifications of the main experiment instruments are listed in Table 1.

2.3. Experiment results

2.3.1. Temperature and humidity distribution

Fig. 5 shows the IAT and OAT measurement results along the height direction of the racks. It is observed that the IAT difference for all racks was relatively small, the maximum of which was 1.6 °C. The average IAT for the racks was 19.2 °C, which was within the recommended range of ASHRAE (DB, 18 °C–27 °C) and far below its upper limit. This means that there was a significant energy saving potential. However, the RH of 68.5% exceeded the recommended upper limit of ASHRAE (RH, 60%), and there was a risk of affecting the normal operation of the servers, such as tape media errors and corrosion. There was an apparent difference in the OAT at different heights of the racks, and almost all racks presented a higher outlet temperature at the top region than at the bottom region. This can be attributed to the fact that the hot air exhausted from the racks flows upward with buoyancy. According to Meng et al. [42], the large difference between the OAT and IAT of the ITE can reflect a high heat dissipation. The temperature difference of row C of 8.3 °C exceeded that of row D of 6.8 °C, meaning that the heat



Fig. 3. Measurement point of each CRAC outlet.

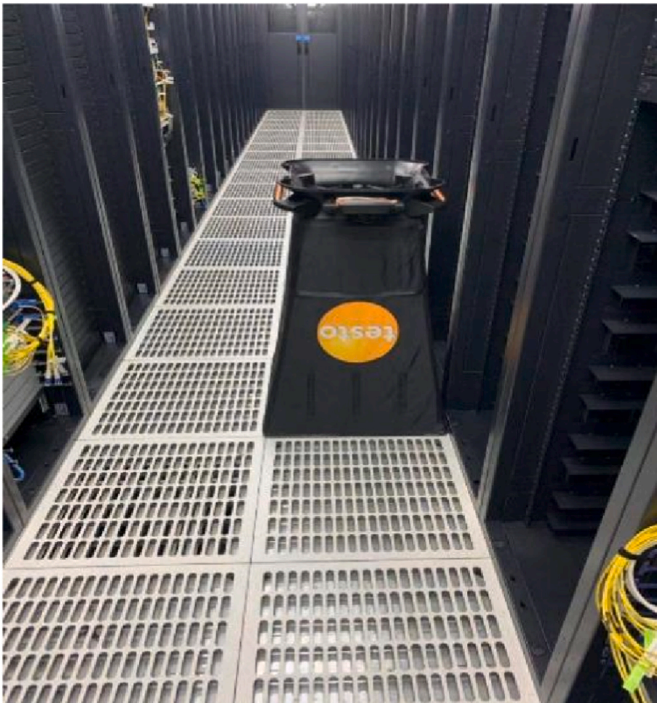


Fig. 4. Field measurement of perforated tiles.

dissipation of row C was higher than that of row D. This is consistent with the fact that the actual server power of row C was larger than that of row D. The main field-testing parameters of the CRACs are presented in Table 2. It can be seen that there was an obvious difference in both the SAT and RAT for each CRAC, as well as the fan speed. This is because the

Table 1

Specifications of main experiment instruments.

Experimental instruments	Model	Measuring range	Accuracy
Temperature-humidity recorder	HOBO UX100-011	−20 °C–70 °C, 1%–95% RH	±0.21 °C (0 °C–50 °C) ±2.5% (10%–90%)
Hot-wire anemometer	Testo 425	0–20 m/s	±(0.03 m/s + 5% measurement value)
Flow hood	Testo 420	40–4000 m ³ /h, −20 °C–70 °C	±3% measurement value
Infrared thermal imager	FLIR T530	−20 °C–120 °C	±0.5 °C (0 °C–70 °C) ±0.8 °C (−20 °C–0 °C) ±2 °C

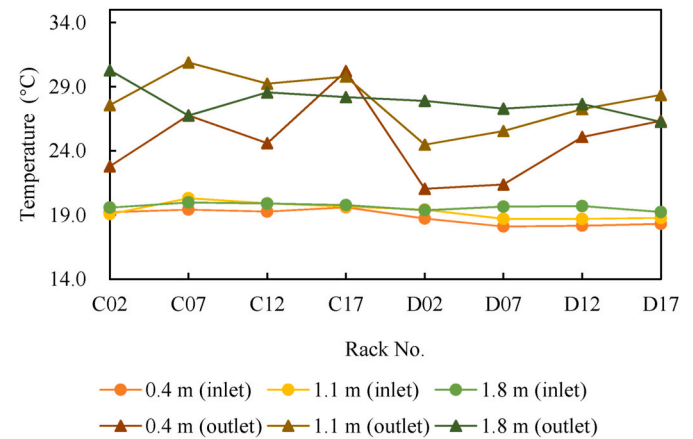


Fig. 5. Measured temperature of racks along the height direction.

Table 2

The main field-testing parameters of each CRAC.

CRAC No.	1#	2#	3#	4#	5#	6#	mean
SAT (°C)	22.0	17.1	20.3	14.8	14.8	closed	17.8
RAT (°C)	27.3	26.7	25.4	22.1	21.2		24.5
Percentage of fan speed (%)	100	79	70	86	86		84

servers in the DC were not uniformly arranged, and the servers concentrated in racks A–D accounted for up to 90% of the total power, resulting in each CRAC running under different cooling loads. Moreover, the average difference between the RAT and SAT of the CRACs at 6.7 °C was lower than the recommendation of the Chinese *Code for the Design of Data Centers* (8 °C–15 °C), leading to an increase in supply air volume and thus larger power consumption of the fans. According to the measurement results, the fitted curves of the fan speed in terms of the air volume and power consumption were extracted, as demonstrated in Fig. 6. From that, the actual air volume as a function of the fan speed presented a linear relationship. Further, the actual fan power consumption and fan speed satisfied the power function model, and the fitting exponent was 2.784 power, which was somewhat different from the theoretical value of 3 power. The goodness of fit R^2 of the above-mentioned two fitted curves were as high as 0.989 and 0.995, respectively. Hence, the total supply air volume and electricity consumption of the CRACs were calculated to be 148385.76 m³/h and 36 kW, respectively.

2.3.2. Airflow and thermal performance evaluation

According to field observations, except for some of the racks in rows C and D, no voids were installed with blanking panels, which easily

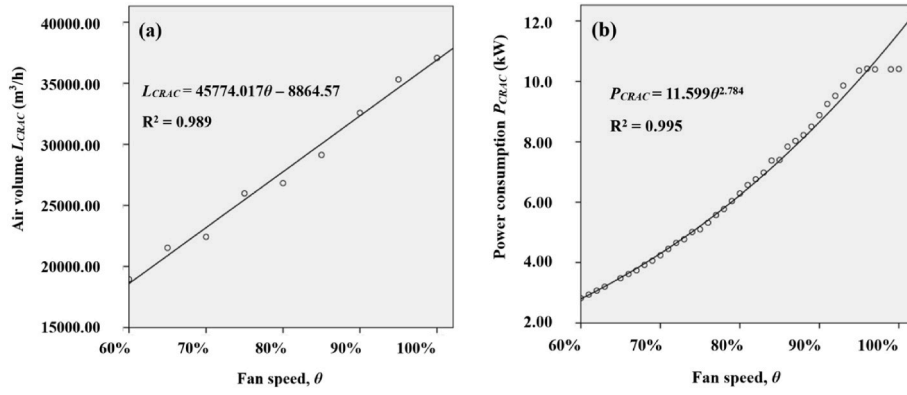


Fig. 6. Fitted functions to the measured parameters of CRACs: (a) Air volume vs. fan speed and (b) Fan power consumption vs. fan speed.

Table 3

SHI and RTI of the DC based on field experiments.

Rack No.	SHI	RTI	Rack No.	SHI	RTI
C02		98.78%	D02		137.17%
C07	0.207	88.23%	D07		122.80%
C12		93.32%	D12	0.177	93.09%
C17		74.79%	D17		88.44%

caused the hot-cold air mixing through voids. Unnecessary perforated tiles corresponding to empty racks were not closed, leading to part of the cold air not cooling the servers but directly returning to CRACs through empty racks, and the ineffective air supply volume of the CRACs increasing.

To quantitatively assess the airflow and thermal performance within the DC, the supply heat index (SHI) [49] and return temperature index (RTI) [50] were introduced to estimate the extent of hot-cold air mixing and the cooling condition on rack level. SHI reflects the effect of hot air exhausted from the ITE on the SAT of the CRAC, the formula of which is shown in Eq. (1). Here, SHI = 0 indicates an ideal air distribution, and in general, SHI < 0.2 indicates a good airflow condition. The closer the SHI approaches zero, the less the mixing extent of the hot and cold airflows is. RTI reflects the airflow management level around the rack, which corresponds to the extent of cold air bypassing and hot air recirculation, the formula of which is shown in Eq. (2). In addition, RTI = 100% and RTI = 95%–105% are regarded as the ideal and good airflow distribution, respectively. RTI < 100% indicates bypass and RTI > 100% indicates recirculation. The more these values deviate from 100%, the more severe the problem.

$$SHI = \frac{\sum_i \sum_j (T_{in,i,j}^r - T_{sup}^c)}{\sum_i \sum_j (T_{out,i,j}^r - T_{sup}^c)} \quad (1)$$

$$RTI = \frac{T_{ret}^c - T_{sup}^c}{T_{out}^r - T_{in}^r} \times 100\% \quad (2)$$

where T_{in}^r and T_{out}^r are the IAT and OAT of the rack, respectively. The subscript i indicates the i th rack in a row, and j indicates the j th row in DC. In addition, T_{sup}^c and T_{ret}^c are the SAT and RAT of the CRAC, respectively.

Table 3 presents the SHI and RTI of the DC based on the field experiments. The SHI of rows C and D were approximately 0.2, which signifies that a certain hot-cold air mixing occurred. This demonstrates that even when cold aisles were fully contained, the hot and cold air were not sufficiently separated. This is mainly caused by the fact that blanking panels were not installed in the voids (especially on the top region) of all racks in rows C and D. The SHI of row C was larger than that of row D, which is mainly because there was higher server power in row C than that in row D, which led to the OAT of row C higher. The RTI reached up to 137.17%, which can be ascribed to the low server power and the air mixing through voids, resulting in a small difference between IAT and OAT and thus a high RTI value. Meanwhile, the RTI in some racks ranged 74.79%–93.32% (e.g., C07, C17, D12, and D17), which shows that the cold air bypass was serious. In other words, cold air from the CRACs could not remove the heat generated by the servers, but rather directly returned to the CRACs, leading to a higher supply air volume than that required by the servers, resulting in energy inefficiency. Fig. 7 presents the infrared thermography images of rows A and C. It can be seen that the local overheating phenomenon existed at

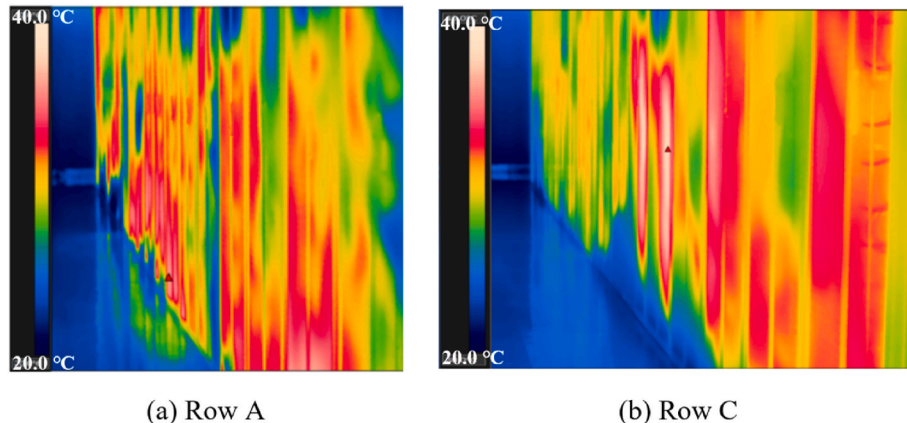


Fig. 7. Infrared thermography images of rack outlet in rows A and C.

Table 4
Thermal environment optimization schemes.

Optimization schemes	Enclosing the cold aisles	Closing unnecessary perforated tiles	Installing blanking panels	Setting RAT of CRACs
Scheme 1 (benchmark)	✓			field experiment data
Scheme 2	✓	✓		
Scheme 3	✓		✓	
Scheme 4	✓	✓	✓	
Scheme 5	✓	✓	✓	26 °C
Scheme 6	✓	✓	✓	27 °C
Scheme 7	✓	✓	✓	28 °C

Note: The software 6SigmaRoom can set the RAT directly.

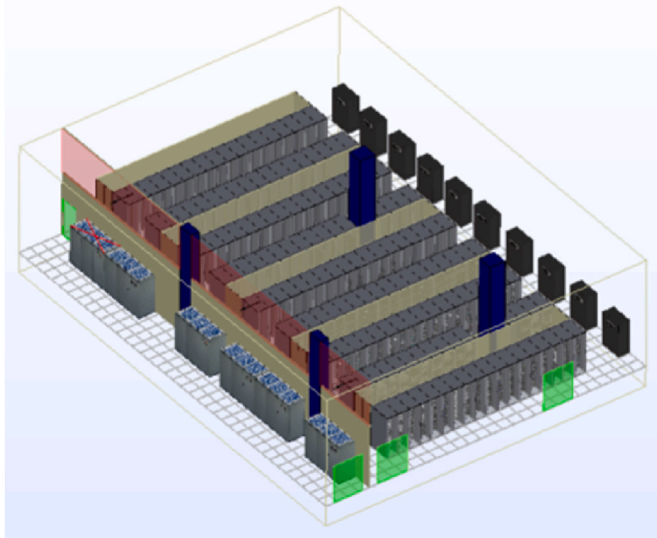


Fig. 8. Model of the studied DC.

the rack outlet of row A and row C, and this issue was more severe for row A in which none of the voids were installed with blanking panels. It is found that the local temperature of the rack outlets reached 40 °C, which may threaten the normal operation of the servers.

The above analysis indicates that the RH of the rack inlet air

Table 5
The measured power for every rack in the DC (unit: kW).

Rack No.	Row A	Row B	Row C	Row D	Row E	Row F	Row G	Row H	Row I
1	3.92	3.90	3.63	2.52	0.00	0.00	0.00	0.00	0.00
2	3.84	3.88	3.84	2.55	0.00	0.00	0.00	0.00	0.00
3	3.00	3.87	3.71	2.58	0.00	0.00	0.00	0.00	0.00
4	3.24	3.90	3.40	2.67	0.00	0.00	0.00	0.00	0.00
5	3.31	3.70	3.57	2.73	0.00	0.00	0.00	0.00	0.00
6	3.17	3.74	3.61	2.74	0.00	0.00	0.00	0.00	0.00
7	3.09	3.84	3.34	2.77	0.00	0.00	0.00	0.00	0.00
8	3.70	3.79	3.35	2.76	1.03	1.15	0.00	0.00	0.00
9	3.13	3.80	3.49	2.81	2.17	1.89	0.00	0.00	0.00
10	2.93	3.73	3.57	2.77	1.84	0.00	0.00	0.00	0.00
11	3.17	3.80	3.61	2.64	1.02	1.13	0.00	0.00	0.00
12	3.17	0.00	3.66	2.76	1.73	1.95	0.00	0.00	0.00
13	2.98	3.70	2.39	2.75	2.06	0.00	0.00	0.00	0.00
14	3.17	3.70	2.62	2.75	0.00	0.00	0.00	0.00	0.00
15	3.14	3.81	3.26	3.64	0.00	0.00	0.00	0.00	0.00
16	3.39		3.86	3.67	0.00	0.00	0.00	0.00	0.00
17	3.26		3.55	3.57	0.00		0.00	0.00	0.00
18	3.13		3.85	3.53	0.00		0.00	0.00	0.00

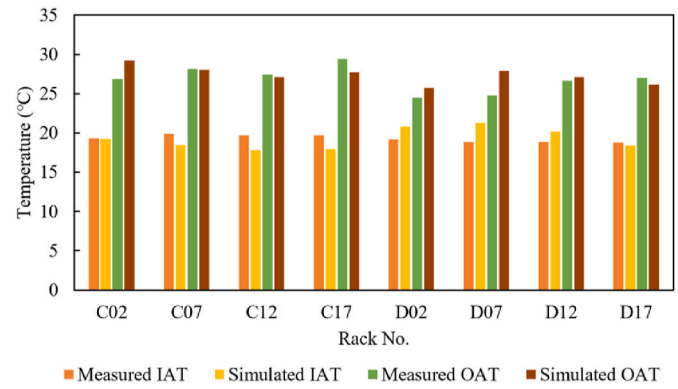


Fig. 9. Comparison between measured and simulated values.

exceeded the recommended range of ASHRAE, the mixing of hot and cold airflows, local overheating, and cold energy waste in the studied DC. Therefore, the thermal environment needs to be improved.

3. Optimization and prediction of thermal environment

3.1. Optimization schemes

To address the above-mentioned problems, three relatively cost-effective and convenient optimization methods were proposed, including installing blanking panels at all voids of the racks, closing unnecessary perforated tiles corresponding to empty racks, and increasing the set RAT of the CRACs. Considering that the design air temperature in the hot aisles was 28 ± 2 °C, three settings of RAT (26 °C, 27 °C, and 28 °C) were chosen to determine the most reasonable value. These can be combined into seven optimization schemes, as listed in Table 4. Scheme 1 represents the original DC with only the cold aisles contained, which was used as the benchmark.

3.2. Effects prediction

The software 6SigmaRoom was applied to numerical simulations on the effects of various optimization methods. The model established according to the actual geometry and configuration is shown in Fig. 8. The settings and data inputs, including the server power, operating parameters of the CRACs, open area of the perforated tiles and ceiling grilles, correlative temperature sensors, and other factors, were based on field experiments. The measured power consumption of each rack is listed in Table 5, and the power and layout of servers in the racks were consistent

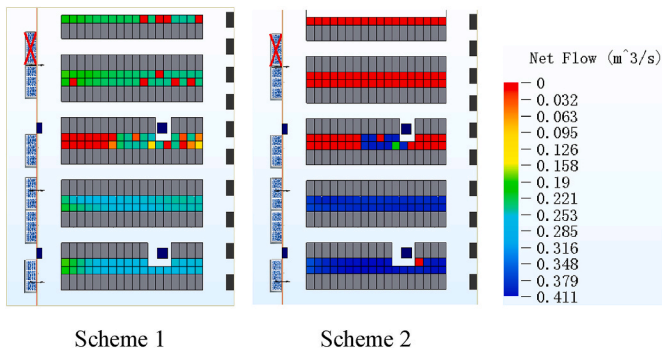


Fig. 10. Airflow rate of perforated tiles of Scheme 1 and Scheme 2.

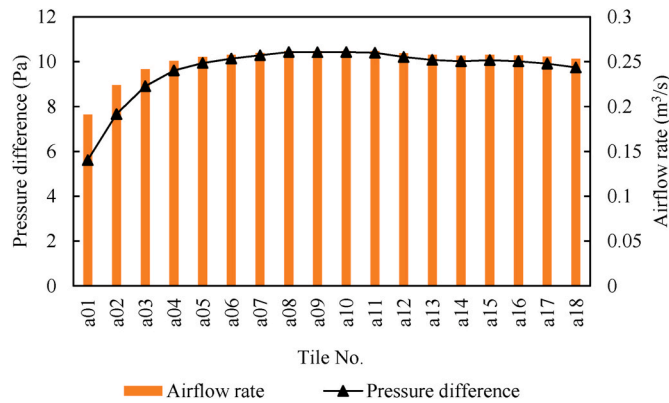


Fig. 11. The pressure difference and airflow rate through perforated tiles of Row A.

with field experiments. The user could choose the servers from the vendor libraries of 6SigmaRoom according to the brand model. And the server model was a black box that needed to define its power consumption. Besides, the simulations adopted the standard $k-\epsilon$ turbulence model. The structured linear solver with the AMG solving format was used for these simulations. The structured grid was employed, and the grid number was set as 7242568 in the simulations to achieve high accuracy. In addition, the disturbance of air leakage and the obstruction in the raised plenum were not considered. A comparison between the simulation and field experiment results for the rack temperature is presented in Fig. 9. As the figure shows, the relative deviation between the simulation and experiment values for the IAT and OAT of the testing racks were all within 12%, and the accuracy of the simulation was validated.

The airflow rate of the perforated tiles in Scheme 1 and Scheme 2 are shown in Fig. 10. From that, there exists invalid supply air across the

perforated tiles of the empty racks in the original DC. Additionally, the airflow rate supplied to the server rack far from the CRACs is higher than that near the CRACs. That is mainly because the static pressure distribution in the plenum underneath the raised floor is uneven. Taking the perforated tiles of row A as an example, as the distance from CRACs increases, the static pressure difference across perforated tiles gradually grows till it reaches the peak value and then decreases slightly (see Fig. 11). It implies that the airflow rate through perforated tiles near CRACs (i.e., a01, a02, and a03) is relatively low. In Scheme 2, the supply air volume of the perforated tiles along the row tends to be more even, and its maximum increases from $0.266 \text{ m}^3/\text{s}$ to $0.411 \text{ m}^3/\text{s}$. Consequently, closing unnecessary perforated tiles can prevent cold air from bypassing through empty racks, and the ineffective air supply of CRACs is significantly reduced.

The IAT of the racks corresponding to Schemes 1–3 is shown in Fig. 12. In Scheme 1, a better thermal environment occurs in rows C–F, and the IAT is between 15°C and 27°C . For rows A and B, however, the IAT of the racks near CRACs exceed 27°C , and half of them exceed the allowable temperature of 32°C in ASHRAE, which is the region where the hotspots are located. However, the IAT of the racks far from CRACs is 18°C – 27°C . It can be concluded that the IAT distribution along the cold aisle is noticeably non-uniform in the original DC. This is because the supply air volume is insufficient for the racks close to CRACs in rows A and B, leading to the strong mixing of hot and cold airflow, which increases the rack IAT. In Scheme 2, benefiting from the increase of the airflow rate through perforated tiles, the overall rack IAT decreases. The rack IATs in Scheme 3 are all below 27°C , which means that the uniformity of rack IAT distribution and the cooling efficiency are improved. In addition, compared with closing unnecessary perforated tiles, installing blanking panels has a more remarkable improvement effect in the thermal environment. This demonstrates that installing blanking panels based on CAC can effectively address the hot-cold air mixing and ensure the working environment of ITE.

Fig. 13 shows the temperature distribution along the height direction in the DC. This signifies that the overall temperature distribution in the original DC is non-uniform. The OAT of the racks rises with the height, and the cooling effects of the racks far from CRACs are better than those near CRACs. In addition, the OAT at the top regions of the racks near CRACs in rows A and B is as high as 43.0°C , indicating that hot spots are particularly obvious. While the air temperature in hot aisles where rows F–I are located is only approximately 15.8°C . There are four reasons for this. First, the servers are unevenly arranged, forming a high heat dissipation region (rows A–D) and a low heat dissipation region (rows E–I). Second, the rack IAT distribution is non-uniform, as shown in Fig. 12, and the cold airflow rate decays continuously as the height increases. Third, the hot air converges on the top of the racks with buoyancy and recirculates to heat the rack inlet air through the gaps. Finally, as the voids of the racks are not installed with blanking panels and the unnecessary perforated tiles are not closed, the cold air supplied from the CRACs is leaked and bypassed, resulting in cooling energy

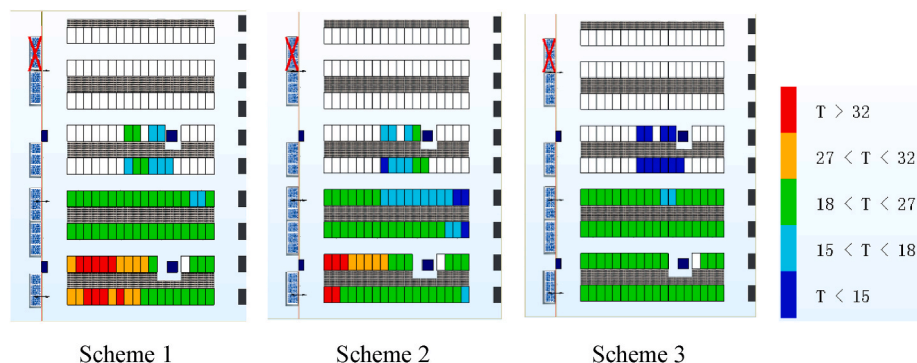


Fig. 12. IAT of racks in Scheme 1, Scheme 2, and Scheme 3.

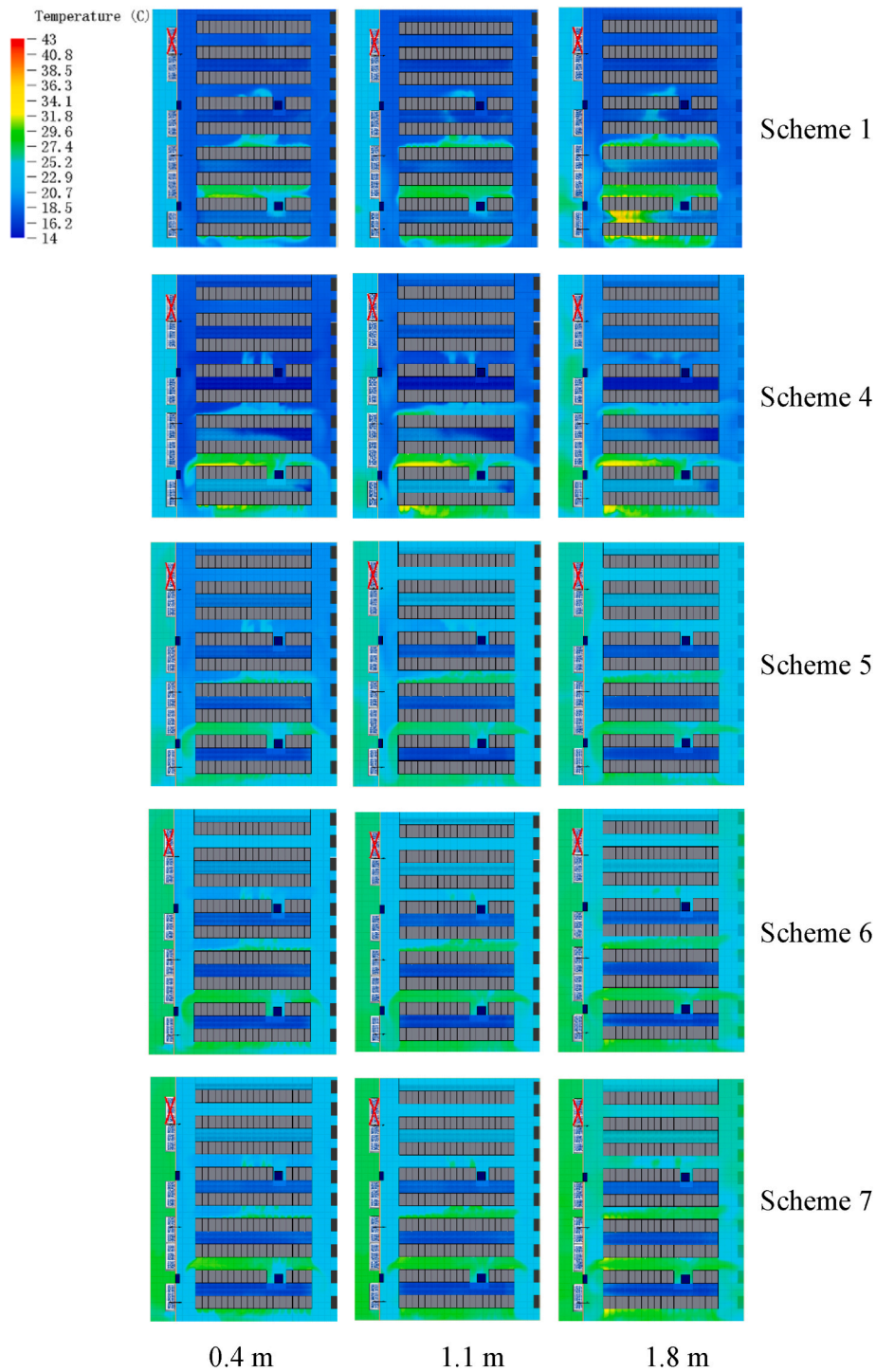


Fig. 13. Temperature field of optimization schemes along the height direction.

waste. When combining the installation of blanking panels with the closure of unnecessary perforated tiles, the cold air only flows across servers to the adjacent hot aisle, and thus the airflow mixing is significantly reduced, and the cold energy is fully utilized. In Scheme 4, the hotspot temperature in the DC decreases to 32.9 °C with a reduction of 10.1 °C, and the maximum temperature difference decreases by 8.9 °C. The results show that the overall temperature distribution in the DC is more even and hot spots are significantly eliminated. Compared with Scheme 4, Schemes 5–7 present a more uniform thermal distribution and almost the same maximum air temperature. The SHI and RTI corresponding to the optimization schemes of the DC based on numerical

Table 6

SHI and RTI corresponding to optimization schemes of the DC based on numerical simulations.

Index	Scheme 1	Scheme 4	Scheme 5	Scheme 6	Scheme 7
SHI	0.130	0.039	0.030	0.033	0.030
RTI	68.3%	92.7%	93.3%	97.9%	96.6%

simulations are presented in Table 6. In Scheme 4, while the SHI decreases to 0.039, the RTI increases to 92.7%. It is demonstrated that the rack IAT is closer to the SAT of the CRAC, indicating that the cooling

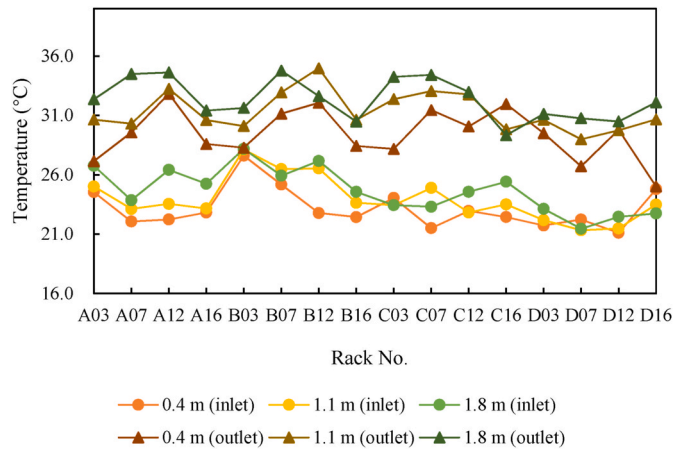


Fig. 14. Measured rack temperature along the height direction after optimization.

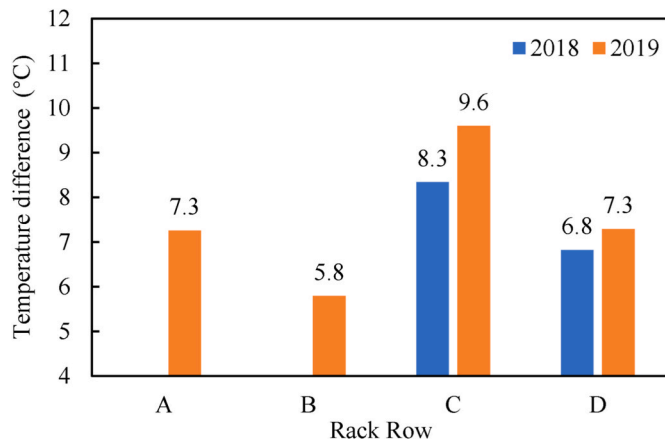


Fig. 15. Average difference between the OAT and IAT of racks before and after optimization.

efficiency is improved and the bypass of cold air is significantly reduced. With the set RAT increasing, the SHI remains nearly stable, whereas the RTI increases to 97.9% and then decreases to 96.6%. The SHI and RTI of Scheme 7 are still better than those of Scheme 4. It is known that the higher the set RAT is, the lower the supply air volume and fan speed of the CRACs. Considering the appropriate thermal environment and low fan power comprehensively, it is reasonable to set the RAT to 28 °C. At the same moisture content, the higher DB temperature results in a lower RH. When adjusting the return air parameter setting to 28 °C DB and 40% RH, the rack inlet air state point is located within the recommended envelope, and the RH is improved.

4. Field experiment and analysis after optimization

4.1. Comparison of thermal environment before and after optimization

Based on the predicted effects of optimization schemes via numerical simulations, optimization scheme 7 was implemented in the studied DC. To examine the application effects of the optimization methods, the field experiment was conducted again during August 11–13, 2019. During the measurement, the total IT load increased to 317.14 kW, and the test rows were newly added to A–D.

4.1.1. Comparison of temperature and humidity distribution

The measured rack temperature along the height direction after optimization is shown in Fig. 14. The IAT of the racks was approximately

Table 7

Main field-testing parameters of CRACs after optimization.

CRAC No.	1#	2#	3#	4#	5#	6#	mean
SAT (°C)	23.2	18.5	20.7	closed	20.3	/	20.7
RAT (°C)	30.8	30.2	27.5		29.6		29.5
Percentage of fan speed (%)	100	100	60		77	60	79

Note:/represents that the SAT and RAT of CRAC No. 6# cannot be obtained owing to the restriction of the site condition.

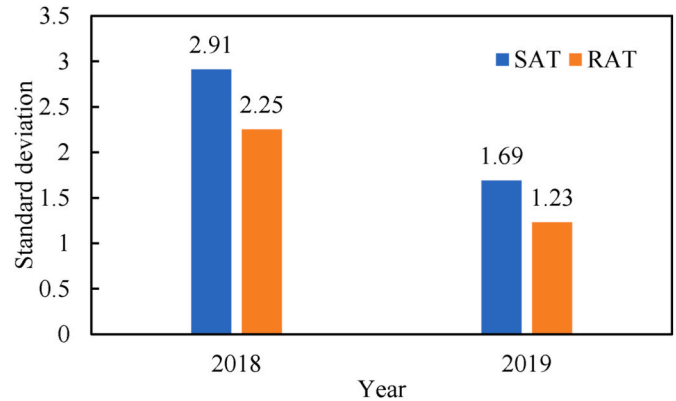


Fig. 16. Standard deviation of the SAT and RAT of CRACs before and after optimization.

among 21.0 °C–27.0 °C and the OAT was among 27.1 °C–35.0 °C, which shows that the thermal environment in the DC was appropriate. However, the IAT of rack B03 exceeded 28 °C, signifying that much of the hot air recirculated into the cold aisle and thus heated the inlet air. The difference between the OAT and IAT was only 2 °C. This is due to the low server power in rack B03. Compared with the original DC, the average IAT and DP temperature of the racks increased to 23.91 °C and 14.5 °C, respectively, and the RH decreased to 54.65%, all within the recommended range of ASHRAE. Fig. 15 shows the average difference between the OAT and IAT of the racks before and after optimization. It can be seen that the temperature difference of row C and row D both rose, which illustrates that the server power of the two rows increased. Table 7 lists the main field-testing parameters of the CRACs after optimization. From that, the mean difference between the RAT and SAT of the CRACs after optimization increased to 8.8 °C within the recommended range of the Chinese Code for the Design of Data Centers. After the calculation, the total supply air volume of the CRACs was reduced to 137482.39 m³/h with a reduction of 10903.37 m³/h, and the total electricity saving of the CRACs fans reached up to 96 kWh/day. Fig. 16 shows that the standard deviation of the SAT and RAT of the CRACs after optimization decreased by 1.22 and 1.02, respectively, which can be conducive to enhancing the temperature distribution uniformity within the DC.

4.1.2. Comparison of airflow and thermal performance

Table 8 shows the SHI and RTI of the DC after optimization. As the table indicates, the middle racks performed better as compared to both end racks of the row. This can be ascribed to the fact that the average difference between the OAT and IAT of the middle racks was higher than that of both end racks, as shown in Fig. 14. The RTI of the racks corresponding to that in the 2018 measurement (such as C07, D07, and D12) were closer to the ideal condition after optimization. Therefore, the implemented measures achieved the improvement of thermal environment in the original DC. However, the SHI and RTI of some racks are still not ideal. There are three reasons for this finding. First, the total server power increased, and it failed to install blanking panels in the voids of racks in time during the frequent addition and removal of the servers,

Table 8

SHI and RTI of the DC based on field experiments after optimization.

Rack No.	SHI	RTI	Rack No.	SHI	RTI	Rack No.	SHI	RTI	Rack No.	SHI	RTI
A03	0.296	205.97%	B03	0.457	474.53%	C03	0.322	116.54%	D03	0.194	99.02%
A07		112.56%	B07		135.54%	C07		95.23%	D07		111.80%
A12		99.94%	B12		124.53%	C12		109.17%	D12		96.02%
A16		146.94%	B16		154.72%	C16		140.75%	D16		143.08%

resulting in the hot-cold air mixing. Second, the servers were unevenly arranged, forming a more obvious high heat dissipation region and a low heat dissipation region compared with that in 2018. Finally, during the 2019 measurement, the standby CRAC (No. 4#) was closer to the high heat dissipation region compared with that during the 2018 measurement (No. 6#), which resulted in a large amount of hot air being more easily drawn into the cold aisle and heating the server inlet air. Therefore, optimization measures need to be further put into practice to achieve the expected results. In addition, the servers should be arranged as evenly as possible, and it is necessary to reduce the deterioration of the thermal environment caused by the unreasonable location of standby CRACs.

4.2. Further works

There are some limitations in this paper. To ensure the normal and uninterrupted operation of the DC, we focused more on cost-efficient and easy-to-implement optimization approaches. However, the control strategy significantly influences the thermal environment in DCs. As the total power of ITE increases and its arrangement varies, the impact of the standby CRAC location on the thermal environment becomes highlighted, and the corresponding optimization measures require further research. Further, more scrutiny is needed to determine a rational control strategy for the DC. Because when the CRACs adopt the mode of patrol switching and independent return air control, the ITE are unevenly distributed, each CRAC runs under different cooling loads, resulting in cold energy waste and a negative effect on the thermal performance. There is an approach to remove these deficiencies. The group control strategy for controlling CRACs under a coordinated state can address the problems of disorderly competition and energy waste caused by the independent control while improving the thermal environment. Additionally, the appropriate application of the group control strategy will be the focus of future research works.

5. Conclusion

The thermal environment of an existing DC was evaluated through field experiments. Cost-efficient and easily implemented approaches for improving the thermal environment were then proposed. The most effective optimization scheme was determined by predicting and comparing their effects through numerical simulations. Finally, field experiments were conducted again to examine the practical application effects after implementing the optimization scheme. The following conclusions were drawn.

- (1) It is found that there were some thermal environment problems in the original DC, including hot-cold air mixing, a local overheating, a small difference between the RAT and SAT of the CRACs, and an RH of the rack inlet air of 68.5%, exceeding the recommended range of ASHRAE.
- (2) By combining the installation of blanking panels with the closure of unnecessary perforated tiles, it is revealed that increasing the set RAT to 28 °C would result in optimized results. The simulation results show that the hotspot temperature can be approximately reduced by 10 °C, and the thermal distribution is more uniform. Furthermore, the SHI decreases from 0.130 to 0.030, and RTI is optimized from 68.3% to 96.6%.

- (3) Based on the on-site measurement results, after implementing the optimization scheme, the RH of the rack inlet air decreased to 54.65%, the temperature distribution became more even, and the thermal environment was noticeably improved. Meanwhile, the average difference between the RAT and SAT of the CRACs increased from 6.7 °C to 8.8 °C, reducing the energy consumption of the fans by about 96 kW h/day. In addition, optimization measures should be further put into effect to reach the expected results.
- (4) For existing DCs, the most foundational but effective improvement measures are not taken seriously. The performed case study proves that such measures would be cost-effective. Furthermore, compared with the exploitation of a single improvement measure, the coordinated application of multiple technologies leads to the best results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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